

OPERATION HYDRA

Unraveling a Multi-Vector Cyber Crime Involving Phishing, Spoofing & Financial Fraud

Assignment 2 – Unit 2: Types of Cyber Crimes

B.Tech CSE (Data Science)

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Declaration

This project is an academic simulation. All domains, IP addresses, transaction records, identifiers and malware-analysis artifacts are dummy/safe data. The IP addresses are documentation-range examples and the attachment is non-malicious. No real credentials, payment details, victims or operational attack infrastructure are used.

Abstract

Operation Hydra models a multi-stage cybercrime campaign combining phishing, spoofing, a harmless attachment-analysis exercise and simulated financial layering. The investigation correlates email headers, authentication results, IOCs and dummy transactions while considering legal and ethical issues.

1. Objectives

Decompose the campaign; analyze four simulated phishing emails; document SPF/DKIM/DMARC findings; classify a harmless attachment; trace dummy transactions; map technical activity to Indian and international legal frameworks; and produce reproducible GitHub documentation.

2. Case Scenario

A fictional group called Hydra sends targeted messages impersonating campus IT, cloud billing, scholarship and wallet-rewards teams. The scenario assumes a fictional victim follows a lure. Dummy payment records then show rapid transfers through simulated mule accounts.

3. Attack Timeline

09:14 phishing → 10:02 attachment lure → 11:28 credential lure → 09:36–09:48 simulated transaction layering.

4. Legal Mapping

Phishing, spoofing, unauthorized-access concepts, malware-delivery vectors and financial-fraud simulation may implicate different provisions depending on facts. Indian analysis should consider the IT Act and BNS; international comparison may include the Budapest Convention and CFAA concepts. Academic mapping only.

5. Phishing & Spoofing

Four synthetic headers demonstrate SPF/DKIM/DMARC findings, timestamps and documentation-range source IPs.

6. Malware Analysis

No functional malware is included. The HTML artifact is harmless and is hashed by the supplied analysis script.

7. Financial Fraud Tracing

The dummy flow USER-104 → MULE-201 → MULE-202 → WALLET-SIM-01 illustrates rapid simulated layering.

8. Evidence Handling

Preserve originals, hash evidence, analyze working copies, record timestamps and document transformations.

9. Ethics & Impact

Privacy, authorization and proportionality must be considered in real investigations. Synthetic data minimizes risk.

10. Recommendations

MFA, password managers, domain verification, DMARC enforcement, transaction monitoring, awareness training and coordinated response.

11. Conclusion

The project demonstrates correlation of email, endpoint and financial artifacts while maintaining a safe academic environment.